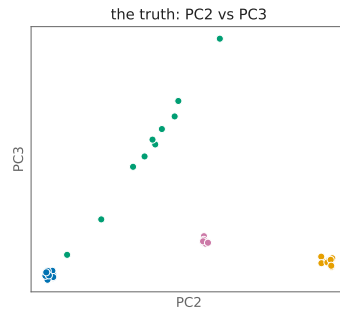
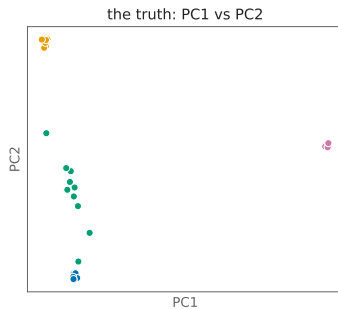
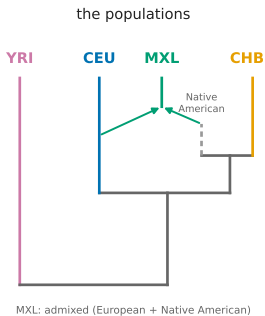
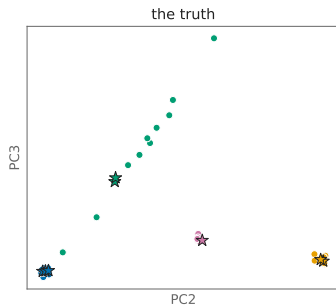
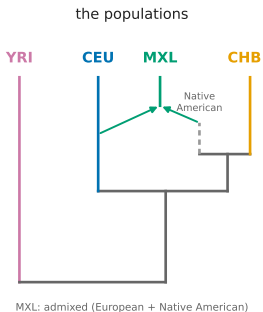




● CEU ● CHB ● MXL ● YRI

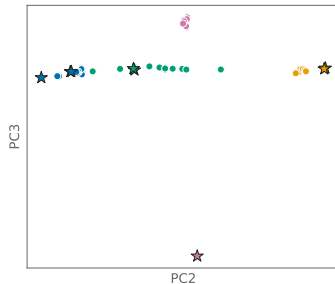
The truth: a PCA of 326 other admixTjeck2 individuals, with these 40 projected onto it.

Four populations — this is what PCA should find.

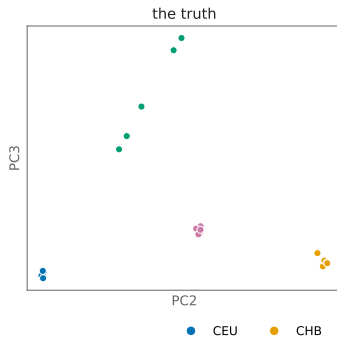
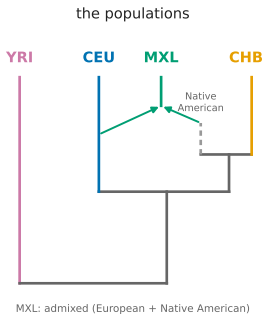


● CEU ● CHB ● MXL ● YRI ☆ relative

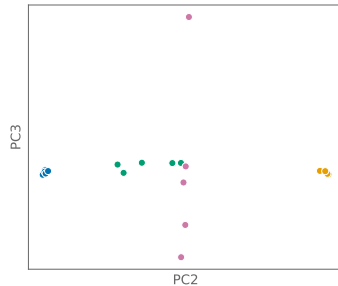
standard PCA, 10 per population + 8 relatives
PC3 recovered: $R^2 = 0.01$



8 relatives among 40 people — and one ancestry axis is gone.



standard PCA, 5 per population, no relatives
PC3 recovered: $R^2 = 0.02$



5 people per population, no relatives — and PCA fails again.

Relatedness in PCA

and how to deal with it

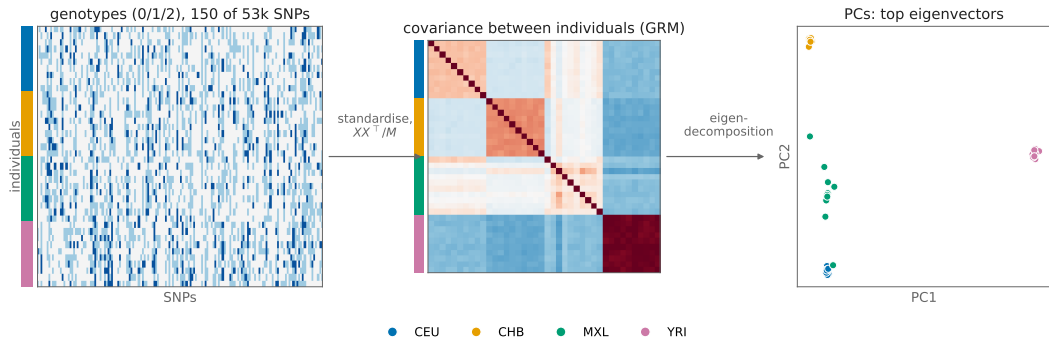
relatePCA

October 5, 2026

The goal

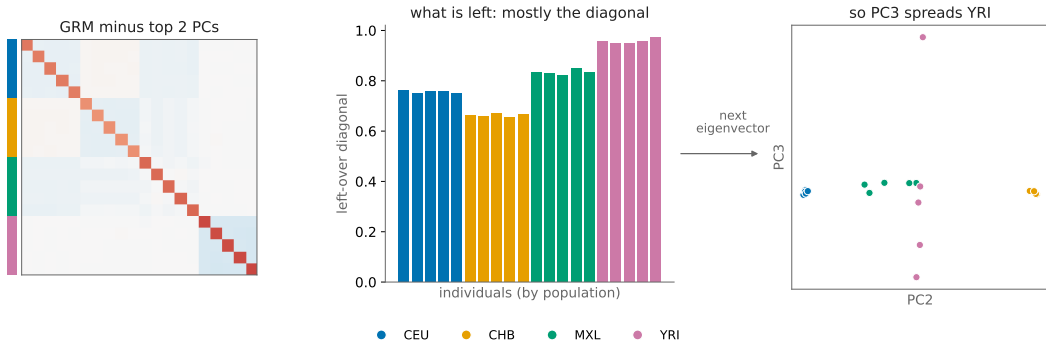
- PCs that reflect **ancestry** — not families, not heterozygosity.
- **Everyone in-sample**: nothing dropped, nothing projected.
- From **20 to 100,000** individuals.
- And estimate the relatedness itself — also for **admixed** people.

PCA in one picture



The PCs are the main axes of the covariance between individuals.

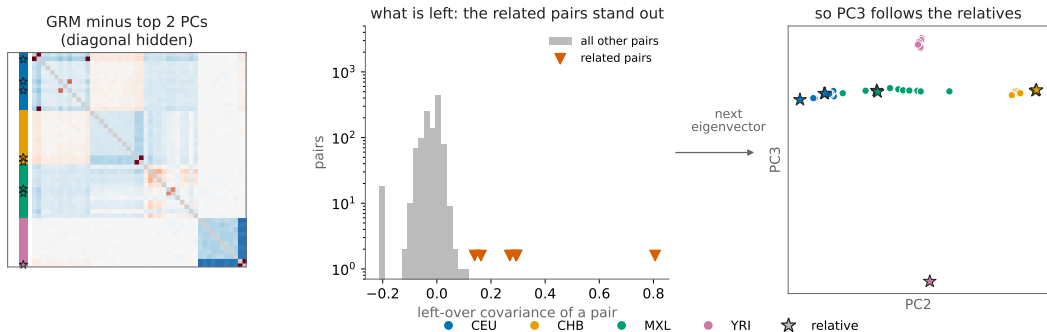
What the top two PCs leave unexplained



5 per population, no relatives. 84% of what is left sits on the diagonal; PC3 puts 99% of its weight on YRI.

YRI heterozygosity is the largest thing left — so PC3 models it.

...and with relatives

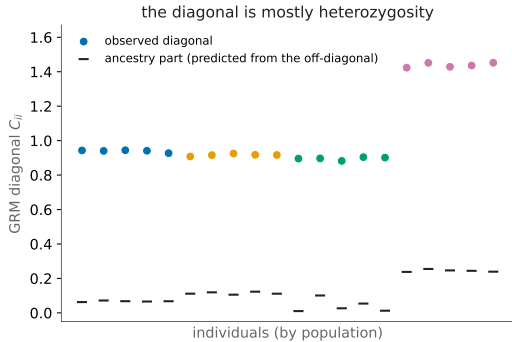
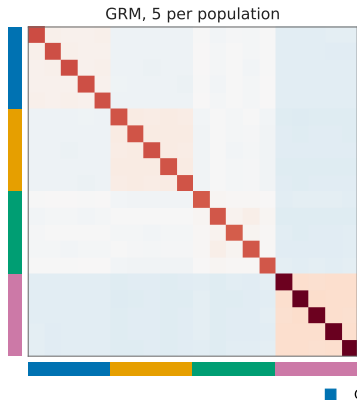


10 per population + 8 relatives. Related pairs: 0.33 on average vs. a spread of 0.04 for all other pairs; PC3 puts 77% of its weight on the 11 individuals in related pairs.

Off the diagonal, relatives dominate — so PC3 models them.

The small- N problem

The GRM diagonal



- Diagonal = ancestry part + heterozygosity noise — and the noise differs by population.
- PCA spends an axis on it, **even without relatives**.

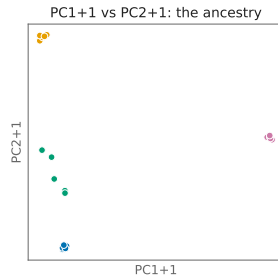
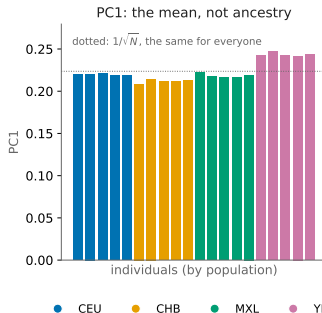
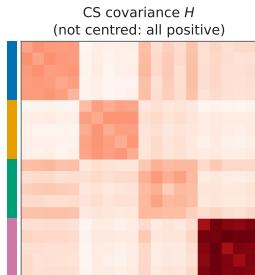
The methods

Small N : fix the diagonal — three options

1. **Impute it**: predict the diagonal from the off-diagonal entries; the rest goes into S (`--impute-diag`).
2. **Remove the heterozygosity** from it: Chen & Storey (next slide).
3. **Whiten**: scale each person by their own noise, so the noise is the same for everyone.

Remove the heterozygosity: Chen & Storey

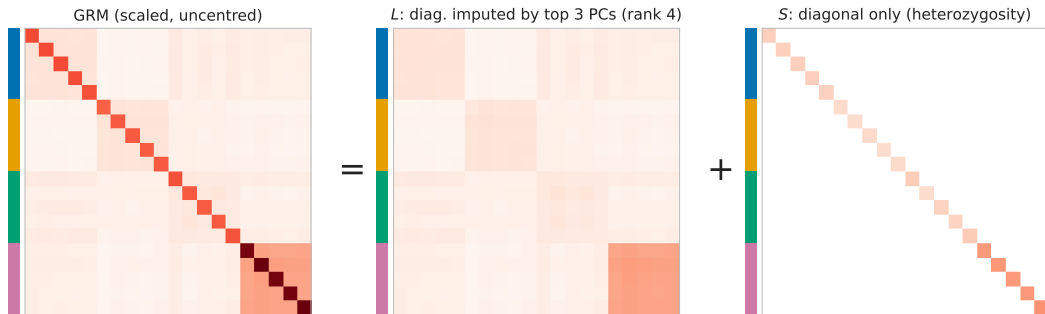
$$H = \frac{1}{M} GG^{\top} - \text{diag}(h_i), \quad h_i = \frac{1}{M} \sum_s g_{is}(2 - g_{is})$$



G not centred; h_i : individual i 's heterozygosity. 5 per population, no relatives.

Not centred: PC1 is the mean — ancestry starts at PC1+1.

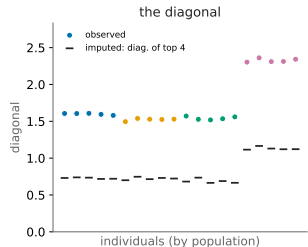
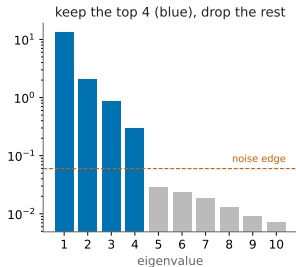
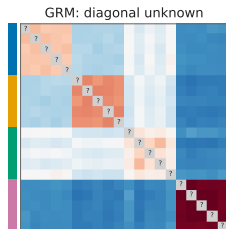
Small N as $L + S$ — robust PCA



5 per population, no relatives. With relatives, S will also hold the related pairs.
Two ways to fit it: principal component pursuit (PCP; aarobust-kin) or a fixed-rank fit (the other three).
Uncentred matrices: PC1 is the mean, so we write PC_{k+1} for the PC that corresponds to standard PC_k .

L : GRM, diagonal imputed from the top PCs. S : the rest.

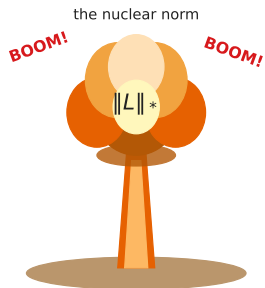
Imputing the diagonal (1): the top r PCs



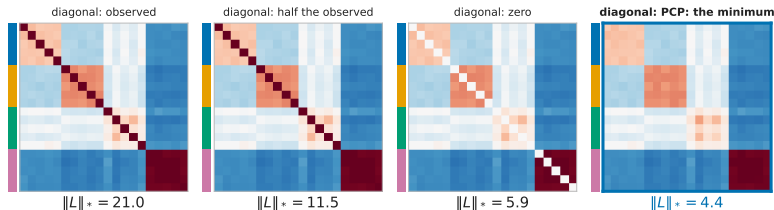
1. Guess the diagonal (e.g. the row means).
2. Keep the top r PCs: $L_r = V_{[:,1:r]} \Lambda_{[1:r,1:r]} V_{[:,1:r]}^T$ (eigenvectors V , eigenvalues Λ).
3. Set the diagonal to the prediction: $A_{ii} \leftarrow (L_r)_{ii} = \sum_{k=1}^r \lambda_k V_{ik}^2$.
4. Repeat until it stops changing.

r grows while the next eigenvalue is above the noise edge. Used by the uncentred GRM fit, detect-white and dwg.

Imputing the diagonal (2): the nuclear norm



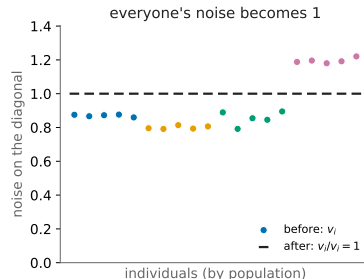
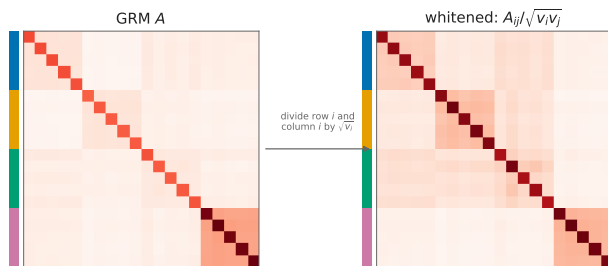
the same GRM off the diagonal, four diagonals: $\|L\|_* = \sum_k |\lambda_k|$



$$\min_{\text{diagonal}} \|L\|_* = \sum_k |\lambda_k| \quad \text{with } L_{ij} = C_{ij} \text{ for } i \neq j$$

No rank to choose; used by PCP (aarobust-kin). Every eigenvector depends on the diagonal, so all of them take part.

What is whitening?

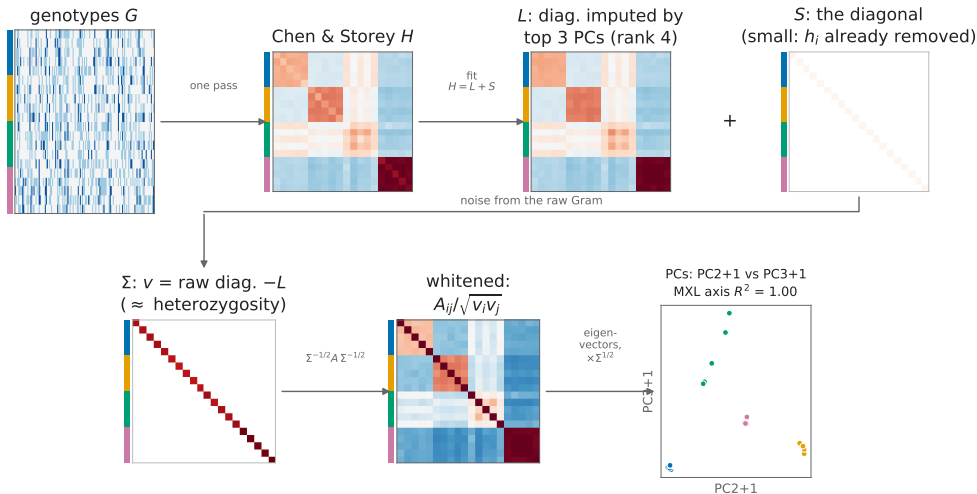


$$\Sigma = \text{diag}(v_1, \dots, v_N), \quad \text{whitened matrix} = \Sigma^{-1/2} A \Sigma^{-1/2}, \quad \text{entries } A_{ij}/\sqrt{v_i v_j}$$

v_i : person i 's noise (the leftover diagonal). After whitening the noise is the same for everyone, and a constant diagonal does not change the eigenvectors. PCs: eigenvectors of the whitened matrix, multiplied back by $\Sigma^{1/2}$.

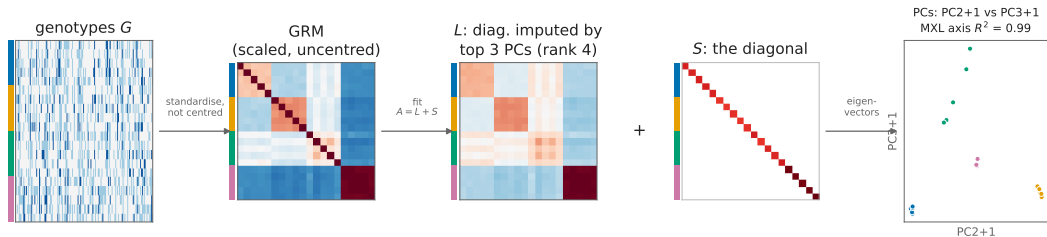
With relatives, Σ will also hold the related pairs.

detect-white: the Chen & Storey matrix



The same fit on H , then each person is scaled by their own noise.

Uncentred GRM fit

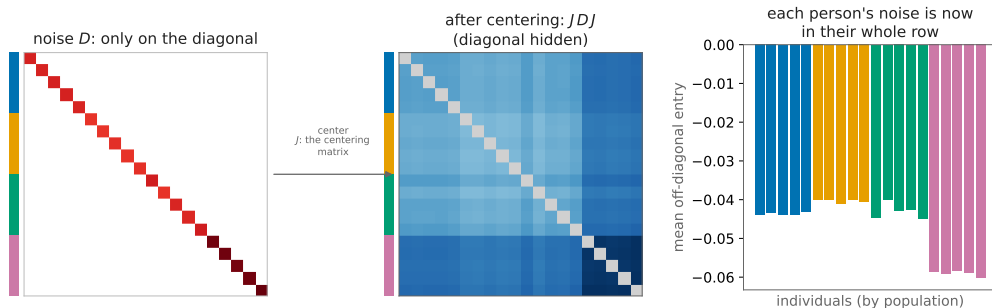


$$\text{Add PC } r+1 \text{ if } \lambda_{r+1}(A - S) > 2\sigma\sqrt{N}, \quad \sigma^2 = \frac{1}{M^2} \sum_s \text{Var}(g_{is}g_{js})/w_s^2$$

σ : the spread of an off-diagonal entry with no structure; $2\sigma\sqrt{N}$: the largest eigenvalue of such noise (Wigner's semicircle, backup). $\text{Var}(g_i g_j) = E[g^2]^2 - (2f)^4$, $E[g^2] = 2f(1-f) + 4f^2$, $w_s = 2f_s(1-f_s)$.

Diagonal free; add imputation PCs while they rise above the noise edge.

A trap: centering

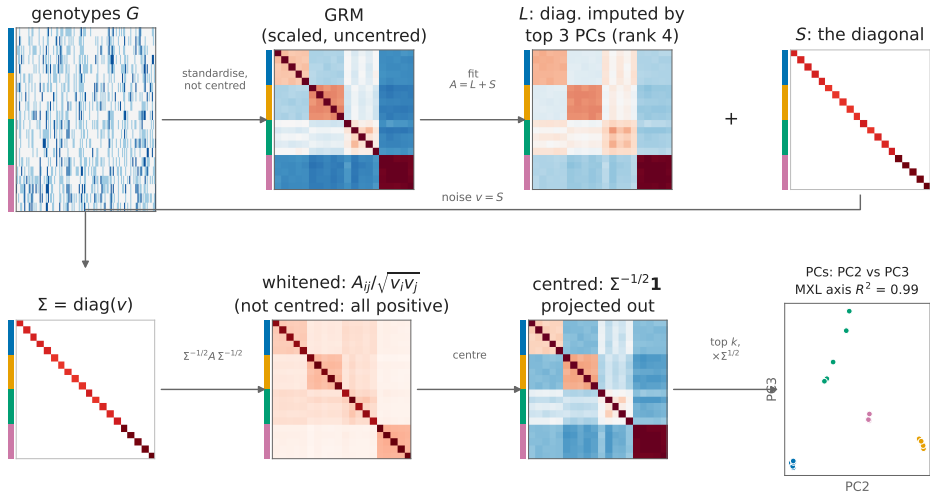


- Centering spreads each person's noise over their whole row and column.
- So freeing the diagonal no longer frees the noise.

Rank-cut fit with the diagonal free, 5 per population: centred, scaled GRM 82% of runs wrong; uncentred, scaled GRM 0%.

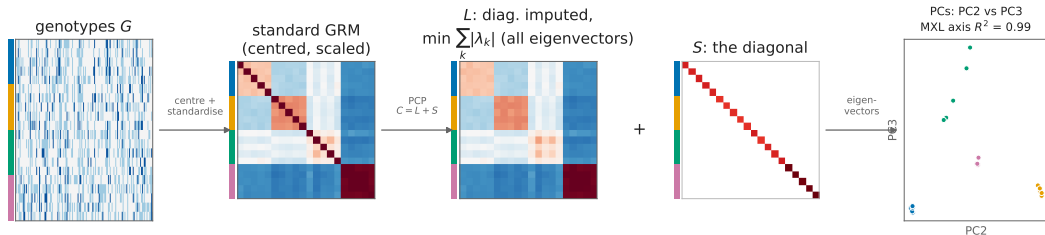
Scaling does not matter. Centering does.

dwg: uncentred GRM, then whitened and centred



The uncentred fit, then whitened and centred — PC1 is ancestry.

aarobust-kin: PCP of the GRM



$L = C$ off the diagonal; its diagonal is the one with the smallest $\sum_k |\lambda_k|$ (nuclear norm). $C = L + S$ exactly, S takes the rest of the diagonal. Each iteration needs a full eigendecomposition.

All eigenvectors, not the top PCs: works centered — but slow.

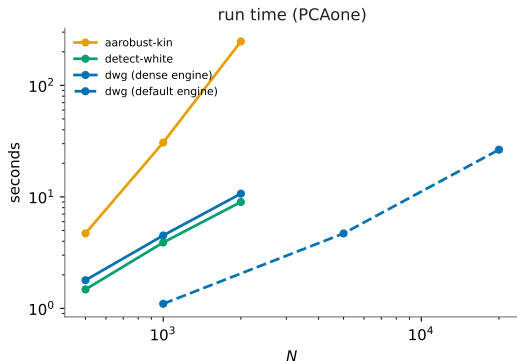
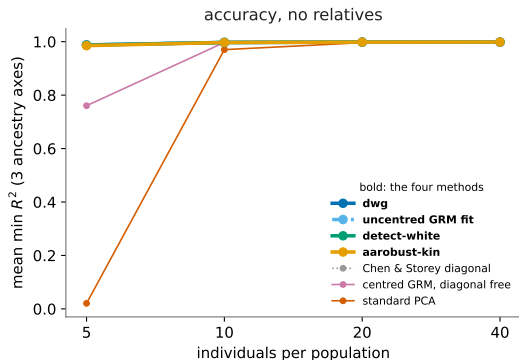
Four ways to fit $L + S$

method	matrix	diagonal imputed from	PCs from
uncentred GRM fit	scaled, uncentred GRM	top r PCs	L
aarobust-kin	centred, scaled GRM	all eigenvectors ($\min \sum \lambda $)	L
detect-white	Chen & Storey (raw)	top r PCs	whitened Gram
dwg	scaled, uncentred GRM	top r PCs	whitened, centred

r : chosen by the noise edge.

All four impute the diagonal and put the rest in S .

Small N : comparing the methods



No relatives, admixTjcek2, 5 replicates per n (numbers in the backup). Run times: PCAone C++.

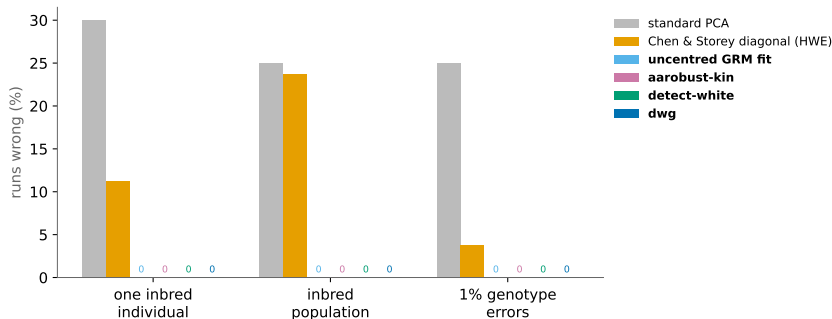
The four are equally accurate — aarobust-kin is the slow one.

Small N : all methods

	mean min R^2 no relatives	needs K ?	top PCs = ancestry?	assumes HWE?	run time $N = 2000$
standard PCA	0.02	yes	yes	no	?
centred GRM, diagonal free	0.76	upper bound	yes	no	?
Chen & Storey diagonal	0.98	yes	no*	yes	?
uncentred GRM fit	0.99	upper bound	no*	no	?
aarobust-kin	0.99	no	yes	no	248 s
detect-white	0.99	upper bound	no*	no	9 s
dwg	0.99	upper bound	yes	no	11 s

5 per population, admixTjeck2, 5 replicates. Run time: PCAone dense engine (dwg in its default engine: 1.1 s at $N = 1000$). ?: not measured. Upper bound: the rank comes from the noise edge, k only caps it. *PC1 is the mean; ancestry starts at PC1+1.

When HWE fails: inbreeding and genotype errors



- Chen & Storey remove the heterozygosity expected under HWE: an inbred person keeps a leftover.
- The four methods leave the diagonal free: **0% wrong** in all three cases.

Runs wrong (min $R^2 < 0.95$), 5–40 per population, 10 replicates each, admixTjeck2.

A free diagonal needs no HWE.

Why relatives break PCA

Relatives share alleles

- PCA looks at how genotypes co-vary between individuals.
- Two sources: shared **ancestry** and recent **relatedness**.

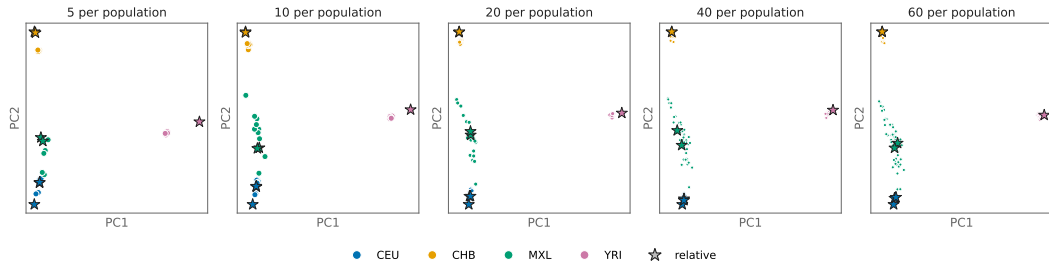
$$E[(g_i - 2\pi)(g_j - 2\pi)] = \underbrace{4(\pi_i - \pi)(\pi_j - \pi)}_{\text{ancestry: few axes} \rightarrow \text{the PCs}} + \underbrace{4\phi_{ij}\sqrt{\pi_i(1-\pi_i)\pi_j(1-\pi_j)}}_{\text{relatedness: should not shape the PCs}}$$

g_i : genotype of individual i at a SNP (0, 1 or 2 copies of an allele); π : allele frequency in the sample; π_i : individual allele frequency, given i 's ancestry;

ϕ_{ij} : kinship given ancestry, the chance that an allele drawn from each is identical by descent ($\frac{1}{4}$ parent-offspring, $\frac{1}{8}$ 2nd degree, 0 unrelated).

- A family is a small, tight cluster — PCA spends components on it.

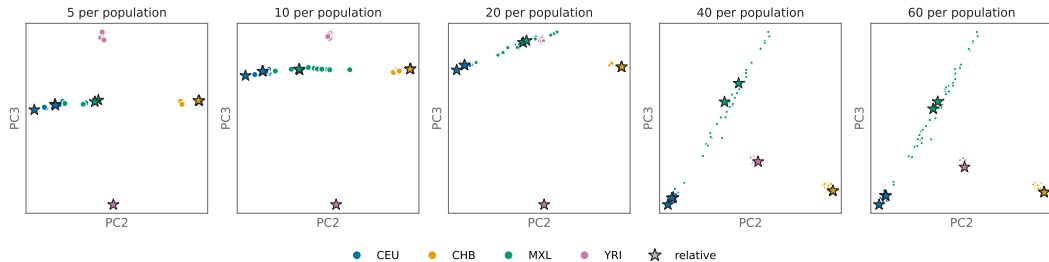
Growing N : PC1 vs PC2



Real genotypes (admixTjeck2); truth: PCA of all 366 individuals.

The same 8 relatives, more and more unrelated individuals.

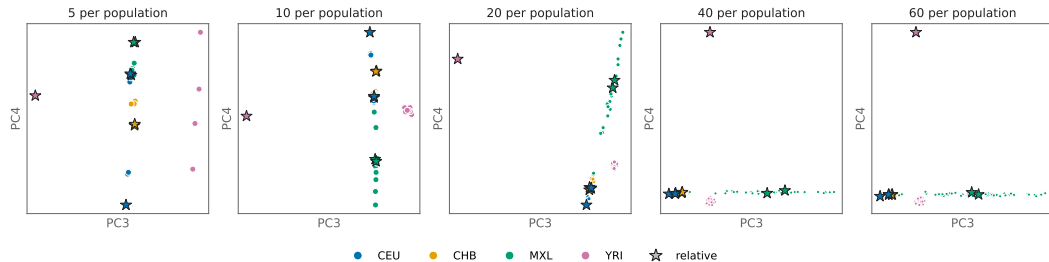
Growing N : PC2 vs PC3



PC3 recovered (R^2): 0.04, 0.01, 0.20, 0.999, 1.00.

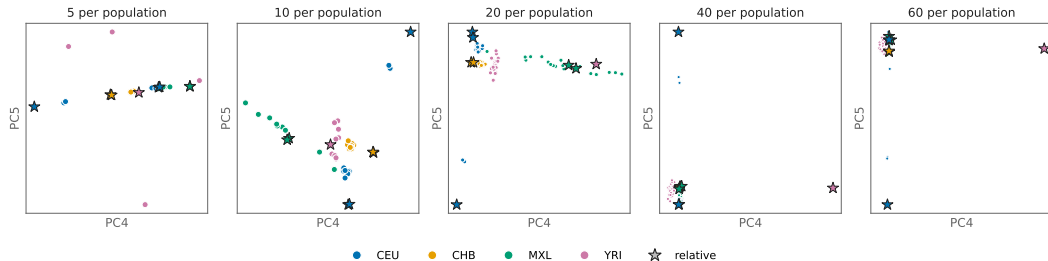
Up to 20 per population PC3 is lost; from 40 it is back.

Growing N : PC3 vs PC4



The relatives do not go away — they move to higher PCs.

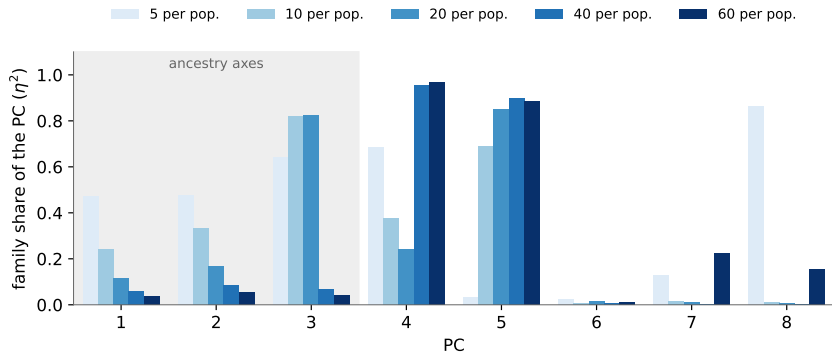
Growing N : PC4 vs PC5



At 20 per population MXL appears on PC4: the ancestry axis the relatives pushed out of PC3.

At large N , PC4 and PC5 each belong to one family.

Growing N : family share of the PCs

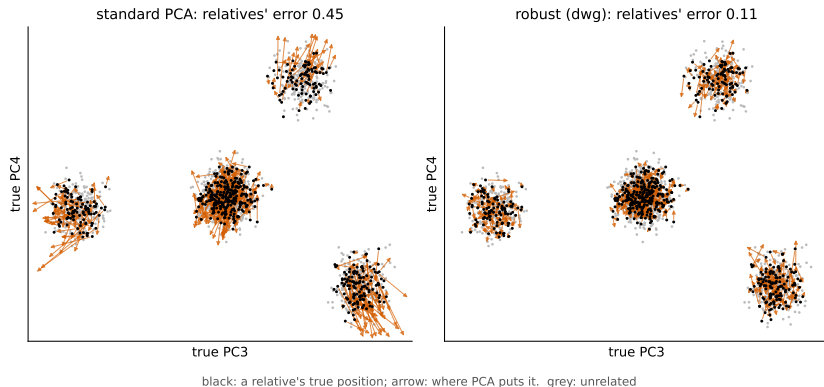


PC4: YRI MZ pair;

PC5: CEU parent-offspring. PC6 is YRI heterozygosity, not a family. Weaker relatives (sibs, 2nd degree) sit at the level of that noise: at 60 per population the CHB sibs are smeared over PC7–8.

Larger N pushes the family axes past the ancestry axes.

Large samples: relatives are misplaced

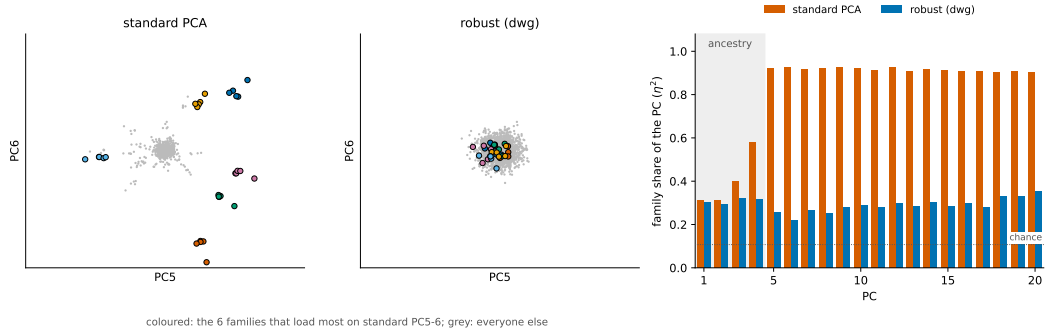


Simulated:

$N = 2000$, 30% in families, 5 populations + an admixed group.

The ancestry axes survive — the relatives move.

Large samples: family axes



Every PC beyond the ancestry axes is a family axis.

The usual fixes

- **Drop relatives** — you lose people.
- **Project relatives** (PC-AiR) — they are no longer in-sample.
- **Screen with KING** — biased for admixed people (later).

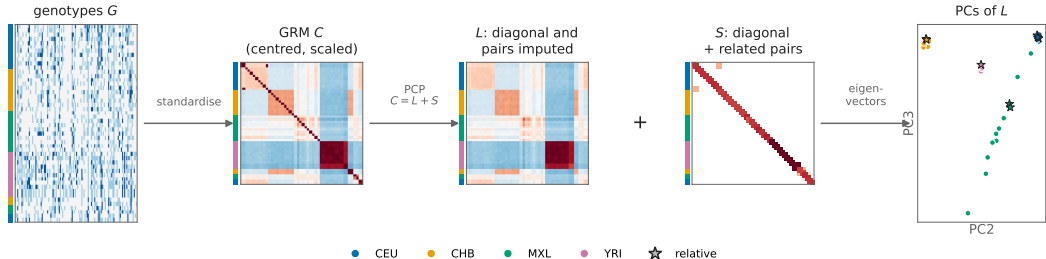
We want to keep everyone in the PCA.

The methods: relatives

The idea

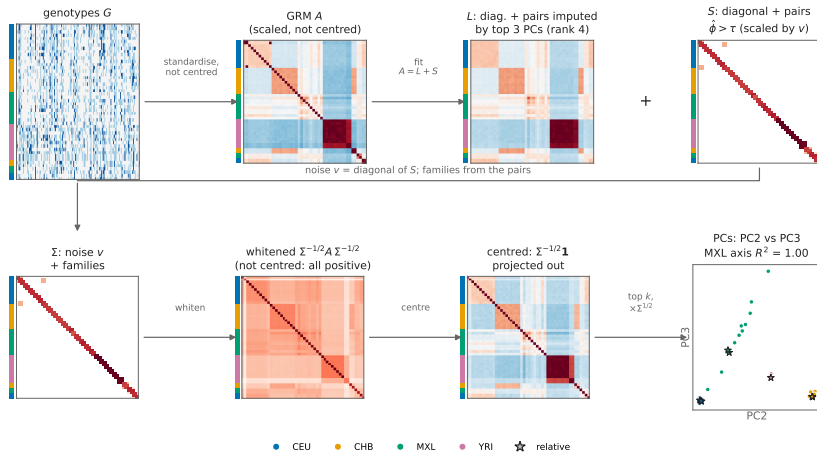
- Split the matrix: **structure** (low rank) + **related pairs** (sparse).
- A pair is related if its kinship $> \tau = 2^{-3.5} \approx 0.088$ (between 2nd and 3rd degree).
- The diagonal stays free.

Robust PCA of the GRM (aarobust-kin)



Accurate, but slow: a full eigendecomposition every iteration.

Detect, then whiten and centre (dwg, the default)



Fast: only the top few eigenvectors. No HWE assumption.

How many PCs? The data decide

- Add axes while they rise above the **noise edge**.
- An axis resting on 2–3 people is a **family**, not ancestry.
- So $-k$ is only an upper bound.

runs that change between $k = 3$ and $k = 10$	old	dwg
of 240	12	2

Whiten — don't just remove the pairs

- Removing the related entries still counts each sibling fully.
- Whitening counts a family by its *effective* size.

family signal in the extra PCs	median η^2
standard PCA	0.92
related pairs imputed	0.37
whitening	0.21

Kinship for admixed individuals

KING is not structure-free

- Correct only for two people from the **same unadmixed** population.
- Relatives of different ancestry are pulled **down**.

true 2nd-degree pair (kinship 0.125)	KING
YRI grandparent – $\frac{1}{4}$ -YRI grandchild	−0.013
CHB aunt – CHB/YRI nephew	0.017

Both are missed by a KING screen.

Use individual allele frequencies

- Predict each genotype from the person's own ancestry: π_{is} .
- What is left over is relatedness.
- PC-Relate, PCAngsd and evalAdmix: **one estimator**.

Use $K - 1$ PCs. More PCs absorb the relatives.

The same pairs, with ancestry taken into account

true 2nd-degree pair (0.125)	KING	evalAdmix-type
YRI grandparent – $\frac{1}{4}$ -YRI grandchild	−0.013	0.085
CHB aunt – CHB/YRI nephew	0.017	0.089

The dwg screen

1. First fit with KING.
2. evalAdmix kinship from the **ancestry axes only**.
3. Confirm with **IBD**: unrelated $k_0 \approx 1$, 2nd degree ≈ 0.5 .

relatives found	KING screen	dwg screen
5–40 per population	86–90%	95–100%

Not just kinship: (k_0, k_1, k_2)

- Parent–offspring and full sibs both have kinship $\frac{1}{4}$.
- k_0 tells them apart: 0 vs $\frac{1}{4}$.

estimated, $N = 2000$	k_0	k_1	k_2
parent–offspring	0.00	1.00	0.00
full sibs	0.25	0.50	0.25
2nd degree	0.50	0.49	0.01

Two details that matter

- Keep each person's own frequencies, $\pi_i \neq \pi_j$.
Pooling them calls a parent–offspring pair 83% unrelated.
- Leave the pair's **whole family** out of the frequency fit.

Does it work?

Small N with relatives

	runs wrong	relatives found	relatives' error
per population:	5 / 10 / 20 / 40	5 / 10 / 20 / 40	5
standard PCA	100 / 60 / 30 / 0%	—	0.40
aarobust-kin	2 / 0 / 0 / 0%	88 / 85 / 82 / 88%	0.14
detect-white	0 / 0 / 0 / 0%	85 / 85 / 77 / 87%	0.13
dwg	0 / 0 / 0 / 0%	94 / 95 / 96 / 100%	0.11

admixTjcek2, 10 scenarios incl. relatives of different ancestry, 5 replicates each; no false pairs for any method.

Relatives found: true pairs with pedigree kinship $\geq \tau$.

All three fix the PCs; dwg finds the most relatives.

Realistic LD: HAPNEST, $N = 5000$

	standard	aarobust-kin	detect-white	dwg
relatives' placement error	0.048	–	0.028	0.019
relatives found	–	–	97%	98%
cross-ancestry grandparents	–	–	66%	87%
time	14 s	too slow	100 s	21 s

aarobust-kin needs a full eigendecomposition per iteration: about an hour per run at $N = 5000$, not run.

The missed pairs share less than expected: realised kinship $< \tau$.

Real panel: 1000 Genomes with ASW

5–20 per population	PCA wrong	relatives found
standard PCA	up to 27%	–
aarobust-kin	0%	78–84%
detect-white	0%	78–83%
dwg	0%	90–96%

CEU, CHB, YRI, ASW; 8 scenarios incl. relatives of different ancestry, 3 replicates each. The few extra pairs are real relatives in the panel.

Bonus: it found real relatives in 1000 Genomes.

Scaling up

Large N : no $N \times N$ matrix

16 threads, in memory	standard PCA	dwg
$N = 5000$	3.5 s	4.7 s
$N = 20,000$	24 s	26.5 s

Robust PCA at about the cost of standard PCA.

Very large N : find relatives with a sketch

- Comparing all pairs is N^2 — too slow at biobank scale.
- Sketch each genome; check only each person's nearest neighbours.

$N = 100,000$	all pairs	sketch
time to find relatives	162 s	52 s
related pairs found	27,880	27,880

Genotype likelihoods (-G)

- Low depth shrinks genotypes towards the mean.
- Undo the shrinkage; the extra noise lands on the free diagonal.

PCA wrong at 4×	10 per pop.	5 per pop.
standard PCA on likelihoods	100%	100%
dwg	0%	12.5%

Try it

```
PCAone -b data -k 10 --robust          # any N; -k is an upper bound  
PCAone -G data.beagle.gz -k 5 --robust # genotype likelihoods
```

- `.eigvecs`: the robust PCs.
- `.relpairs`: related pairs with kinship and (k_0, k_1, k_2) .

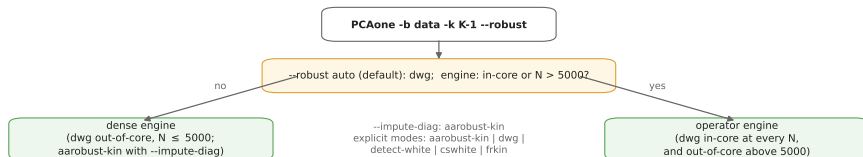
github.com/aalbrechtsen/PCAone, branch `robust-relatives`, docs/`robust/`

Take home

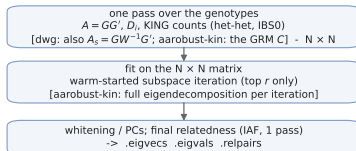
1. Relatives break PCA: a lost axis at small N , family axes at large N .
2. At small N the diagonal breaks it too: work uncentred, diagonal free.
3. Detect the related pairs, then whiten: everyone stays in-sample.
4. For admixed relatives use ancestry-aware kinship, not KING.

Open: missing genotypes; first cousins at ~ 5 per population.

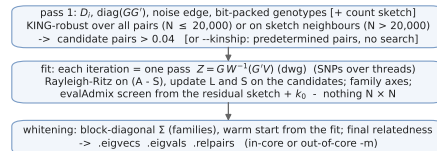
Backup: the engines



dense engine (aarobust-kin always; dwg out-of-core and others for $N \leq 5000$)



operator engine (dwg in-core; other modes $N > 5000$)



Backup: formulas

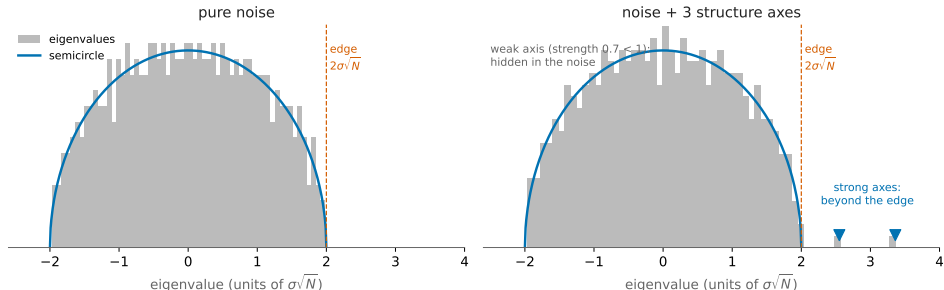
- Kinship from the residual: $\hat{\phi}_{ij} = R_{ij}/(2\sqrt{v_i v_j})$, $R = A - L$, v_i the fitted noise.
- Whitening: $\Sigma = v^{1/2}(I + 2\Psi)v^{1/2}$; PCs of $\Sigma^{-1/2}A\Sigma^{-1/2}$, mapped back by $\Sigma^{1/2}$.
- Centring: $JDJ = D - (d\mathbf{1}' + \mathbf{1}d')/N + \bar{d}\mathbf{1}\mathbf{1}'/N$.
- IBD likelihood: IBD0 $g_i \sim \text{HWE}(\pi_i)$, $g_j \sim \text{HWE}(\pi_j)$; IBD1 one shared allele; IBD2 $g_i = g_j$.

Backup: small N without relatives

mean min R^2 , per population:	5	10	20	40
dwg	0.988	0.997	0.998	0.999
uncentred GRM fit	0.988	0.997	0.998	0.999
detect-white	0.987	0.997	0.998	0.998
aarobust-kin	0.985	0.997	0.998	0.999
Chen & Storey diagonal	0.984	0.996	0.998	0.999
centred, scaled GRM, diagonal free	0.76	0.997	0.998	0.999
standard PCA	0.02	0.97	0.996	0.998

admixTjeck2, 5 replicates per n . The differences among the first five are within noise.

Backup: Wigner's semicircle and the noise edge



- A symmetric $N \times N$ matrix of independent noise (mean 0, spread σ): its eigenvalues fill a semicircle on $[-2\sigma\sqrt{N}, 2\sigma\sqrt{N}]$ (Wigner, 1955).
- So with no structure, no eigenvalue goes beyond $2\sigma\sqrt{N}$: the **noise edge**.
- A structure axis of strength θ shows up beyond the edge only if $\theta > \sigma\sqrt{N}$ (at $\theta + \sigma^2 N/\theta$); weaker axes are lost in the noise.

For the GRM, σ is the spread of an off-diagonal entry for unrelated individuals; the diagonal is in S . Linked SNPs make the real noise somewhat larger, so the edge is approximate.